

Westchester County
Department of Planning
Sidewalk Coverage Analysis

COMPLETE DATA PACKAGE

Geospatial Data, Statistical Analysis,
and Implementation Guidance

502 Priority Gaps • 4,386 Roads Analyzed • \$125.3M Investment

START HERE

Your Complete Guide to the Data Package

Prepared for:

Interested Parties and GIS Integration Teams

October 2025

Quick Access:

GEOSPATIAL_DATA/ → 40+ GeoJSON files
STATISTICAL_SUMMARIES/ → Complete analysis results
LATEX_REPORTS/ → 4 comprehensive PDF reports
ANALYSIS_CODE/ → Sections 10.2, 10.3, 10.4

1 QUICK START: What You Need to Know

1.1 Package Overview

This comprehensive data package contains everything requested for GIS integration and additional analysis of the Westchester County Sidewalk Coverage Analysis.

1.2 Key Deliverables

- **GEOSPATIAL DATA:** 40+ GeoJSON files with complete infrastructure and analysis results
- **STATISTICAL SUMMARIES:** All JSON analysis results and correlations
- **ANALYSIS CODE:** Sections 10.2, 10.3, 10.4 as requested
- **COMPREHENSIVE REPORTS:** 4 detailed LaTeX PDF reports
- **EXCEL DELIVERABLES:** Enhanced analysis workbooks
- **ADDITIONAL MATERIALS:** Tutorials, guides, and documentation

1.3 Key Findings at a Glance

- **4,386 county roads analyzed** for sidewalk coverage
- **Current coverage:** 38.6% countywide, 54.9% in TOD areas
- **502 priority gaps** identified for immediate attention
- **Strong correlations:** Transit proximity (-0.78), property values (+0.71)
- **Economic benefits:** 2.4:1 benefit-cost ratio, 16.6% annual ROI

1.4 Immediate Next Steps

1. **For GIS Integration:** Go to Section ??
2. **For Analysis Code:** See Section ??
3. **For Statistical Data:** See Section ??
4. **For Full Reports:** See Section ??

2 PACKAGE STRUCTURE

2.1 Complete Directory Overview

```
DELIVERABLES_FOR_INTERESTED_PARTY/  
START_HERE.pdf           # This guide  
README.md                # Package documentation  
GEOSPATIAL_DATA/  
  processed/             # All GIS data (40+ files)  
  raw/                   # Analysis results (9 files)  
                           # Original data (22+ files)
```

```

STATISTICAL_SUMMARIES/          # All JSON results (6 files)
ANALYSIS_CODE/                  # Complete analysis scripts
    section_10_2_correlation_analysis.py
    section_10_3_cost_benefit_modeling.py
    section_10_4_implementation_guide.py
LATEX_REPORTS/                  # 4 comprehensive PDF reports
EXCEL_DELIVERABLES/             # Enhanced Excel workbooks
ADDITIONAL_MATERIALS/           # Extras and support materials
ARCHIVE/                         # Source files and backups

```

2.2 File Count Summary

Table 1: Package Contents Summary

Category	File Count
Geospatial Data (GeoJSON)	31+
Statistical Summaries (JSON)	6
Analysis Scripts (Python)	3+
LaTeX Reports (PDF)	4
Excel Workbooks	2+
Documentation Files	5+
Total Files	50+

3 GEOSPATIAL DATA

3.1 What's Included

40+ GeoJSON files organized into:

3.1.1 Processed Analysis Results (9 files)

- `county_wide_coverage.geojson` - Main coverage analysis
- `priority_sidewalk_gaps.geojson` - 502 priority gaps ranked
- `tod_area_roads.geojson` - Transit area analysis
- `metro_north_station_buffers.geojson` - 0.5-mile station buffers

3.1.2 Raw Infrastructure Data (10+ files)

- `westchester_sidewalks.geojson` - Complete sidewalk network
- `westchester_bike_lanes.geojson` - Bicycle infrastructure
- `westchester_metro_north_stations.geojson` - Transit stations

3.2 Technical Specifications

- **Format:** GeoJSON (RFC 7946)
- **Coordinate System:** EPSG:4326 (WGS 84)
- **Recommended for analysis:** EPSG:2263 (NY State Plane)
- **Total size:** 45 MB
- **Compatibility:** QGIS, ArcGIS, Python, R

3.3 Quick GIS Integration

3.3.1 QGIS

1. Open QGIS
2. Drag GeoJSON files into QGIS
3. Reproject to EPSG:2263 for analysis
4. Start analyzing!

3.3.2 Python

```
import geopandas as gpd
```

```
# Load coverage data
```

```
coverage = gpd.read_file('GEOSPATIAL_DATA/processed/county_wide_coverage.geojson')
```

```
gaps = gpd.read_file('GEOSPATIAL_DATA/processed/priority_sidewalk_gaps.geojson')
```

```
# Reproject for analysis
```

```
coverage = coverage.to_crs('EPSG:2263')
```

```
gaps = gaps.to_crs('EPSG:2263')
```

3.4 Documentation

- **Complete GIS guide:** `ADDITIONAL_MATERIALS/GUIDES/gis_integration_tutorial.md`
- **Data dictionary:** `LATEX_REPORTS/geospatial_data_documentation.pdf`

4 STATISTICAL SUMMARIES

4.1 What's Included

6 JSON files with complete statistical analysis:

- `sidewalk_coverage_statistics.json` - Main coverage metrics
- `county_wide_statistics.json` - County-wide analysis
- `tod_statistics.json` - Transit area statistics
- `priority_gaps_list.json` - Complete priority gap rankings

4.2 Key Statistical Findings

Table 2: Key Statistical Results

Metric	Value	Significance
Transit correlation	-0.78	p < 0.001
Property value correlation	+0.71	p < 0.001
TOD coverage advantage	3.01x	p < 0.001
Benefit-cost ratio	2.4:1	87% confidence
Annual ROI	16.6%	Risk-adjusted 14.2%

4.3 Access Methods

4.3.1 Python

```
import json

# Load statistics
with open('STATISTICAL_SUMMARIES/tod_statistics.json', 'r') as f:
    tod_stats = json.load(f)

print(f"TOD coverage: {tod_stats['transit_adjacent_coverage_rate']}%")
```

4.3.2 R

```
library(jsonlite)

# Load statistics
tod_stats <- fromJSON('STATISTICAL_SUMMARIES/tod_statistics.json')
cat("TOD coverage:", tod_stats$transit_adjacent_coverage_rate, "%\n")
```

5 ANALYSIS CODE

5.1 Sections 10.2, 10.3, 10.4 - All Available!

All requested analysis sections are included as complete, executable Python scripts:

5.1.1 Section 10.2: Advanced Statistical Correlation Analysis

- **File:** ANALYSIS_CODE/section_10.2_correlation_analysis.py
- **Content:** Demographic, economic, and geographic correlations
- **Results:** Statistical significance testing and confidence intervals

5.1.2 Section 10.3: Cost-Benefit Investment Modeling

- **File:** ANALYSIS_CODE/section_10_3_cost_benefit_modeling.py
- **Content:** Complete financial modeling and ROI analysis
- **Results:** \$344.8M benefits, 2.4:1 B/C ratio

5.1.3 Section 10.4: Implementation Phasing and Recommendations

- **File:** ANALYSIS_CODE/section_10_4_implementation_guide.py
- **Content:** 10-year implementation roadmap and policy guidance
- **Results:** Four-phase plan with \$125.3M investment strategy

5.2 Running the Analysis

```
cd ANALYSIS_CODE/  
python section_10_2_correlation_analysis.py  
python section_10_3_cost_benefit_modeling.py  
python section_10_4_implementation_guide.py
```

5.3 Requirements

- Python 3.8+
- pandas, numpy, scipy, matplotlib, seaborn
- geopandas (for spatial analysis)

6 COMPREHENSIVE REPORTS

6.1 Four Complete LaTeX PDF Reports

6.1.1 1. Comprehensive Technical Analysis

- **File:** LATEX_REPORTS/comprehensive_technical_analysis.pdf
- **Content:** Complete analysis with Sections 10.2, 10.3, 10.4
- **Length:** 60 pages
- **Includes:** All findings, methodology, results, recommendations

6.1.2 2. Geospatial Data Documentation

- **File:** LATEX_REPORTS/geospatial_data_documentation.pdf
- **Content:** Complete GIS data dictionary and integration guide
- **Length:** 25 pages
- **Includes:** Data schemas, coordinate systems, software instructions

6.1.3 3. Statistical Analysis Report

- **File:** LATEX_REPORTS/statistical_analysis_report.pdf
- **Content:** Complete statistical compendium
- **Length:** 40 pages
- **Includes:** All correlations, significance tests, economic modeling

6.1.4 4. Implementation Guide

- **File:** LATEX_REPORTS/implementation_guide.pdf
- **Content:** Policy recommendations and action plan
- **Length:** 35 pages
- **Includes:** 10-year roadmap, funding strategy, governance

7 ENHANCED EXCEL DELIVERABLES

7.1 Interactive Analysis Workbooks

7.1.1 Enhanced Statistical Analysis

- **File:** EXCEL_DELIVERABLES/enhanced_statistical_analysis.xlsx
- **Features:** Interactive dashboards, pivot tables, charts
- **Content:** All correlations, statistical tests, sensitivity analysis

7.1.2 Comprehensive Cost-Benefit Model

- **File:** EXCEL_DELIVERABLES/comprehensive_cost_benefit_model.xlsx
- **Features:** Adjustable assumptions, scenario analysis, ROI calculations
- **Content:** Complete financial model with sensitivity testing

8 ADDITIONAL MATERIALS

8.1 Presentations and Tutorials

- **Key Findings Presentation:** ADDITIONAL_MATERIALS/PRESENTATIONS/key_findings_summary.md
- **GIS Integration Tutorial:** ADDITIONAL_MATERIALS/GUIDES/gis_integration_tutorial.md
- **Data Attribution Guide:** ADDITIONAL_MATERIALS/DOCUMENTATION/data_attribution_guide.md

8.2 Support Documentation

- Complete data dictionaries and metadata
- Software compatibility guides
- Usage examples and code samples
- Contact information and support resources

9 COMMON USAGE SCENARIOS

9.1 Scenario 1: GIS Integration and Analysis

You want to get the data into GIS for further analysis

Quick Path:

1. Go to `GEOSPATIAL_DATA/`
2. Load `county_wide_coverage.geojson` into your GIS software
3. Load `priority_sidewalk_gaps.geojson` for priority areas
4. Load `westchester_metro_north_stations.geojson` for transit context
5. Start your analysis!

For detailed instructions: See `ADDITIONAL_MATERIALS/GUIDES/gis_integration_tutorial.md`

9.2 Scenario 2: Statistical Analysis

You want to perform additional statistical analysis

Quick Path:

1. Load JSON files from `STATISTICAL_SUMMARIES/`
2. Run analysis scripts from `ANALYSIS_CODE/`
3. Use Excel workbooks in `EXCEL_DELIVERABLES/`
4. Reference full statistical report in `LATEX_REPORTS/`

9.3 Scenario 3: Complete Understanding

You want to understand the full project and findings

Quick Path:

1. Read `LATEX_REPORTS/comprehensive_technical_analysis.pdf`
2. Review key findings presentation in `ADDITIONAL_MATERIALS/PRESENTATIONS/`
3. Explore geospatial data with GIS integration tutorial
4. Contact team for questions (see below)

10 SUPPORT AND CONTACTS

10.1 Technical Questions

- **Email:** gis@westchestergov.com
- **Phone:** (914) 995-4700
- **Department:** Westchester County Planning Department

10.2 Data and Analysis Questions

- **Email:** planning@westchestergov.com
- **Website:** www.westchestergov.com/planning
- **Data Portal:** data.westchestergov.com

10.3 Attribution Required

When using this data, please include the attribution:

"Westchester County Sidewalk Coverage Analysis (2025). Westchester County Department of Planning in partnership with Arcanum Performance Monitoring."

11 QUICK REFERENCE CHEAT SHEET

11.1 File Locations

What You Need	Where to Find It
All GIS data	GEOSPATIAL_DATA/
Statistical results	STATISTICAL_SUMMARIES/
Analysis code	ANALYSIS_CODE/
Complete reports	LATEX_REPORTS/
Excel workbooks	EXCEL_DELIVERABLES/
Tutorials	ADDITIONAL_MATERIALS/GUIDES/

11.2 Key Numbers

- **Roads analyzed:** 4,386
- **Priority gaps:** 502
- **Current coverage:** 38.6% countywide
- **TOD coverage:** 54.9%
- **Total investment:** \$125.3M
- **Benefit-cost ratio:** 2.4:1

11.3 Must-Know Files

- `county_wide_coverage.geojson` - Main analysis results
 - `priority_sidewalk_gaps.geojson` - Priority gaps ranked
 - `section_10_*.py` - Analysis code you requested
 - `comprehensive_technical_analysis.pdf` - Complete findings
-

12 YOU'RE READY!

You now have everything needed to work with the Westchester County Sidewalk Analysis data. The package contains:

- ✓ All geospatial data for GIS integration
- ✓ Complete statistical analysis results
- ✓ Sections 10.2, 10.3, 10.4 analysis code
- ✓ Comprehensive PDF reports
- ✓ Interactive Excel workbooks
- ✓ Tutorials and documentation

Happy analyzing!

Questions? Contact us!

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